



Boot media - manual recovery

Install and maintain

NetApp
June 30, 2026

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Boot media - manual recovery

Boot media manual recovery workflow - AFF A400

The automated recovery of the boot image involves the system automatically identifying and selecting the appropriate boot menu option. It uses the boot image on the partner node to reinstall ONTAP on the replacement boot media in your AFF A400 storage system.

The automated boot media recovery process is supported only in ONTAP 9.17.1 and later. If your storage system is running an earlier version of ONTAP, use the [manual boot recovery procedure](#).

To get started, review the replacement requirements, shut down the controller, replace the boot media, allow the system to restore the image, and verify system functionality.

1

Review the boot media requirements

Review the requirements for replacing the boot media.

2

Check encryption key support and status

Determine whether the system has security key manager enabled or encrypted disks.

3

Shut down the controller

Shut down the controller when you need to replace the boot media.

4

Replace the boot media

Remove the failed boot media from the System Management module and install the replacement boot media, and then transfer an ONTAP image using a USB flash drive.

5

Boot the recovery image

Boot the ONTAP image from the USB drive, restore the file system, and verify the environmental variables.

6

Restore encryption

Restore the onboard key manager configuration or the external key manager from the ONATP boot menu.

7

Return the failed part to NetApp

Return the failed part to NetApp, as described in the RMA instructions shipped with the kit.

Requirements for manual boot media recovery - AFF A400

Before replacing the boot media in your AFF A400 system, ensure you meet the necessary requirements for a successful replacement. This includes making sure you have a USB flash drive with the appropriate amount of storage and verifying that you have the correct replacement boot device.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

NetApp strongly recommends that you enable console logging on the devices you are working with and take the following actions when performing this procedure:



- Leave AutoSupport enabled during your maintenance. This allows NetApp support to receive diagnostic information if issues occur.
- Enable session logging for any CLI sessions. This creates a complete record of all commands and system output during the maintenance activity. For guidance on how to enable session logging, see [How to configure PuTTY for optimal connectivity to ONTAP systems](#).

USB flash drive

- Ensure you have a USB flash drive formatted to FAT32.
- The USB must have sufficient storage capacity to hold the `image_xxx.tgz` file.

File preparation

Copy the `image_xxx.tgz` file to the USB flash drive. This file will be used when you transfer the ONTAP image using the USB flash drive.

From the extracted ONTAP image, copy the `EFI` folder to the root directory of the USB flash drive.

Component replacement

Replace the failed component with the replacement component provided by NetApp.

Controller identification

It is critical to apply the commands to the correct controller when you are replacing the impaired boot media:

- The *impaired controller* is the controller on which you are performing maintenance.
- The *healthy controller* is the HA partner of the impaired controller.

What's next?

After you've reviewed the requirements to replace the boot media, you need to [check encryption key support and status on the boot media](#).

Check encryption key support and status - AFF A400

To ensure data security on your storage system, you need to verify the encryption key support and status on your boot media. Check if your ONTAP version supports NetApp Volume Encryption (NVE), and before you shut down the controller check if the key

manager is active.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

Step 1: Check NVE support and download the correct ONTAP image

Determine whether your ONTAP version supports NetApp Volume Encryption (NVE) so you can download the correct ONTAP image for the boot media replacement.

Steps

1. Check if your ONTAP version supports encryption:

```
version -v
```

If the output includes 1Ono-DARE, NVE is not supported on your cluster version.

2. Download the appropriate ONTAP image based on NVE support:
 - If NVE is supported: Download the ONTAP image with NetApp Volume Encryption
 - If NVE is not supported: Download the ONTAP image without NetApp Volume Encryption



Download the ONTAP image from the NetApp Support Site to your HTTP or FTP server or a local folder. You will need this image file during the boot media replacement procedure.

Step 2: Verify key manager status and back up configuration

Before shutting down the impaired controller, verify the key manager configuration and back up the necessary information.

Steps

1. Determine which key manager is enabled on your system:

ONTAP version	Run this command
ONTAP 9.14.1 or later	security key-manager keystore show • If EKM is enabled, EKM is listed in the command output. • If OKM is enabled, OKM is listed in the command output. • If no key manager is enabled, No key manager keystores configured is listed in the command output.

ONTAP version	Run this command
ONTAP 9.13.1 or earlier	<pre>security key-manager show-key-store</pre> • If EKM is enabled, <code>external</code> is listed in the command output. • If OKM is enabled, <code>onboard</code> is listed in the command output. • If no key manager is enabled, <code>No key managers configured</code> is listed in the command output.

2. Depending on whether a key manager is configured on your system, do one of the following:

If no key manager is configured:

You can safely shut down the impaired controller and proceed to the shutdown procedure.

If a key manager is configured (EKM or OKM):

- a. Enter the following query command to display the status of the authentication keys in your key manager:

```
security key-manager key query
```

- b. Review the output and check the value in the `Restored` column. This column indicates whether the authentication keys for your key manager (either EKM or OKM) have been successfully restored.

3. Complete the appropriate procedure based on your key manager type:

External Key Manager (EKM)

Complete these steps based on the value in the `Restored` column.

If all keys show `true` in the `Restored` column:

You can safely shut down the impaired controller and proceed to the shutdown procedure.

If any keys show a value other than `true` in the `Restored` column:

- a. Restore the external key management authentication keys to all nodes in the cluster:

```
security key-manager external restore
```

If the command fails, contact NetApp Support.

- b. Verify that all authentication keys are restored:

```
security key-manager key query
```

Confirm that the `Restored` column displays `true` for all authentication keys.

- c. If all keys are restored, you can safely shut down the impaired controller and proceed to the shutdown procedure.

Onboard Key Manager (OKM)

Complete these steps based on the value in the `Restored` column.

If all keys show `true` in the `Restored` column:

- a. Back up the OKM information:

- i. Switch to advanced privilege mode:

```
set -priv advanced
```

Enter `y` when prompted to continue.

- ii. Display the key management backup information:

```
security key-manager onboard show-backup
```

- iii. Copy the backup information to a separate file or your log file.

You will need this backup information if you need to manually recover OKM during the replacement procedure.

- iv. Return to admin mode:

```
set -priv admin
```

- b. You can safely shut down the impaired controller and proceed to the shutdown procedure.

If any keys show a value other than `true` in the `Restored` column:

a. Synchronize the onboard key manager:

```
security key-manager onboard sync
```

Enter the 32-character alphanumeric onboard key management passphrase when prompted.



This is the cluster-wide passphrase you created when you initially configured the Onboard Key Manager. If you do not have this passphrase, contact NetApp Support.

b. Verify all authentication keys are restored:

```
security key-manager key query
```

Confirm that the `Restored` column displays `true` for all authentication keys and the `Key Manager type` shows `onboard`.

c. Back up the OKM information:

i. Switch to advanced privilege mode:

```
set -priv advanced
```

Enter `y` when prompted to continue.

ii. Display the key management backup information:

```
security key-manager onboard show-backup
```

iii. Copy the backup information to a separate file or your log file.

You will need this backup information if you need to manually recover OKM during the replacement procedure.

iv. Return to admin mode:

```
set -priv admin
```

d. You can safely shut down the impaired controller and proceed to the shutdown procedure.

Shut down the controller for manual boot media recovery - AFF A400

After completing the NVE or NSE tasks, you need to complete the shutdown of the impaired controller. Shut down or take over the impaired controller using the appropriate procedure for your configuration.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

Option 1: Most configurations

Complete the shutdown of the impaired controller by halting it to the LOADER prompt and capturing the boot environment variables. The LOADER prompt is the safe halted state from which you can replace the boot media.

Steps

- a. Take the impaired controller to the LOADER prompt:

If the impaired controller displays...	Then...
The LOADER prompt	Go to Remove controller module.
Waiting for giveback...	Press Ctrl-C, and then respond <code>y</code> when prompted.
System prompt or password prompt (enter system password)	Take over or halt the impaired controller from the healthy controller: <pre>storage failover takeover -ofnode impaired_node_name</pre> When the impaired controller shows Waiting for giveback..., press Ctrl-C, and then respond <code>y</code>.

- b. From the LOADER prompt, enter: `printenv` to capture all boot environmental variables. Save the output to your log file.



This command may not work if the boot device is corrupted or non-functional.

Option 2: Controller is in a MetroCluster configuration



Do not use this procedure if your system is in a two-node MetroCluster configuration.

Take over and halt the impaired controller so that the healthy controller continues to serve data from the impaired controller's storage. To do this, you suppress automatic case creation in AutoSupport, disable automatic giveback, and bring the impaired controller to the LOADER prompt. The LOADER prompt is the safe halted state from which you can replace the FRU.

- If you have a cluster with more than two nodes, it must be in quorum. If the cluster is not in quorum or a healthy controller shows false for eligibility and health, you must correct the issue before shutting down the impaired controller; see [Synchronize a node with the cluster](#).
- If you have a MetroCluster configuration, you must have confirmed that the MetroCluster Configuration State is configured and that the nodes are in an enabled and normal state (`metrocluster node show`).

Steps

1. If AutoSupport is enabled, suppress automatic case creation by invoking an AutoSupport message:

```
system node autosupport invoke -node * -type all -message MAINT=<number of  
hours down>h
```

This prevents automatic support cases from being opened during your planned maintenance window. The

maximum suppression duration is 72 hours. If your maintenance completes early, you can re-enable case creation by invoking an AutoSupport message with `MAINT=END`. For more information, see [How to suppress automatic case creation during scheduled maintenance windows](#).

The following AutoSupport message suppresses automatic case creation for two hours:

```
cluster1:> system node autosupport invoke -node * -type all -message MAINT=2h
```

2. Disable automatic giveback from the console of the healthy controller: `storage failover modify -node local -auto-giveback false`
3. Take the impaired controller to the LOADER prompt:

If the impaired controller is displaying...	Then...
The LOADER prompt	Go to the next step.
Waiting for giveback...	Press Ctrl-C, and then respond <code>y</code> when prompted.
System prompt or password prompt (enter system password)	Take over or halt the impaired controller from the healthy controller: <code>storage failover takeover -ofnode <i>impaired_node_name</i></code> When the impaired controller shows Waiting for giveback..., press Ctrl-C, and then respond <code>y</code>.

Option 3: Controller is in a two-node Metrocluster

To shut down the impaired controller, you must determine the status of the controller and, if necessary, switch over the controller so that the healthy controller continues to serve data from the impaired controller storage.

About this task

- You must leave the power supplies turned on at the end of this procedure to provide power to the healthy controller.

Steps

1. Check the MetroCluster status to determine whether the impaired controller has automatically switched over to the healthy controller: `metrocluster show`
2. Depending on whether an automatic switchover has occurred, proceed according to the following table:

If the impaired controller...	Then...
Has automatically switched over	Proceed to the next step.
Has not automatically switched over	Perform a planned switchover operation from the healthy controller: <code>metrocluster switchover</code>

If the impaired controller...	Then...
Has not automatically switched over, you attempted switchover with the <code>metrocluster switchover</code> command, and the switchover was vetoed	Review the veto messages and, if possible, resolve the issue and try again. If you are unable to resolve the issue, contact technical support.

3. Resynchronize the data aggregates by running the `metrocluster heal -phase aggregates` command from the surviving cluster.

```
controller_A_1::> metrocluster heal -phase aggregates
[Job 130] Job succeeded: Heal Aggregates is successful.
```

If the healing is vetoed, you have the option of reissuing the `metrocluster heal` command with the `-override-vetoes` parameter. If you use this optional parameter, the system overrides any soft vetoes that prevent the healing operation.

4. Verify that the operation has been completed by using the `metrocluster operation show` command.

```
controller_A_1::> metrocluster operation show
Operation: heal-aggregates
State: successful
Start Time: 7/25/2016 18:45:55
End Time: 7/25/2016 18:45:56
Errors: -
```

5. Check the state of the aggregates by using the `storage aggregate show` command.

```
controller_A_1::> storage aggregate show
Aggregate      Size Available Used% State    #Vols  Nodes      RAID
Status
-----
...
aggr_b2        227.1GB    227.1GB    0% online      0 mcc1-a2
raid_dp, mirrored, normal...
```

6. Heal the root aggregates by using the `metrocluster heal -phase root-aggregates` command.

```
mcc1A::> metrocluster heal -phase root-aggregates
[Job 137] Job succeeded: Heal Root Aggregates is successful
```

If the healing is vetoed, you have the option of reissuing the `metrocluster heal` command with the `-override-vetoes` parameter. If you use this optional parameter, the system overrides any soft vetoes that prevent the healing operation.

7. Verify that the heal operation is complete by using the `metrocluster operation show` command on the destination cluster:

```
mcclA::> metrocluster operation show
  Operation: heal-root-aggregates
    State: successful
  Start Time: 7/29/2016 20:54:41
    End Time: 7/29/2016 20:54:42
    Errors: -
```

8. On the impaired controller module, disconnect the power supplies.

Replace the boot media and prepare for manual boot recovery - AFF A400

To replace the boot media, you must remove the impaired controller module, install the replacement boot media, and transfer the boot image to a USB flash drive.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

Step 1: Remove the controller module

To access components inside the controller module, you must remove the controller module from the chassis.

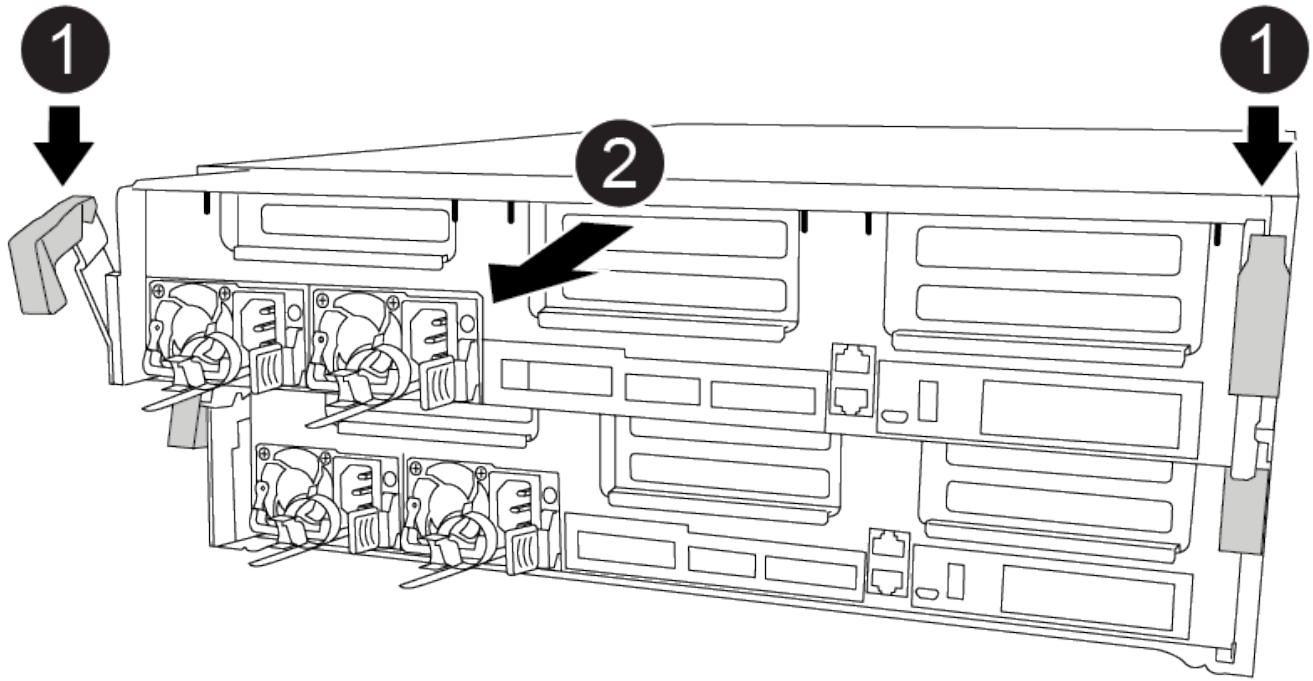
Steps

1. If you are not already grounded, properly ground yourself.
2. Release the power cable retainers, and then unplug the cables from the power supplies.
3. Loosen the hook and loop strap binding the cables to the cable management device, and then unplug the system cables and SFPs (if needed) from the controller module, keeping track of where the cables were connected.

Leave the cables in the cable management device so that when you reinstall the cable management device, the cables are organized.

4. Remove the cable management device from the controller module and set it aside.
5. Press down on both of the locking latches, and then rotate both latches downward at the same time.

The controller module moves slightly out of the chassis.



1	Locking latches
2	Controller moves slightly out of chassis

6. Slide the controller module out of the chassis.

Make sure that you support the bottom of the controller module as you slide it out of the chassis.

7. Place the controller module on a stable, flat surface.

Step 2: Replace the boot media

You must locate the boot media in the controller module (see the FRU map on the controller module), and then follow the directions to replace it.

Before you begin

Although the contents of the boot media is encrypted, it is a best practice to erase the contents of the boot media before replacing it. For more information, see the [Statement of Volatility](#) for your system on the NetApp Support Site.



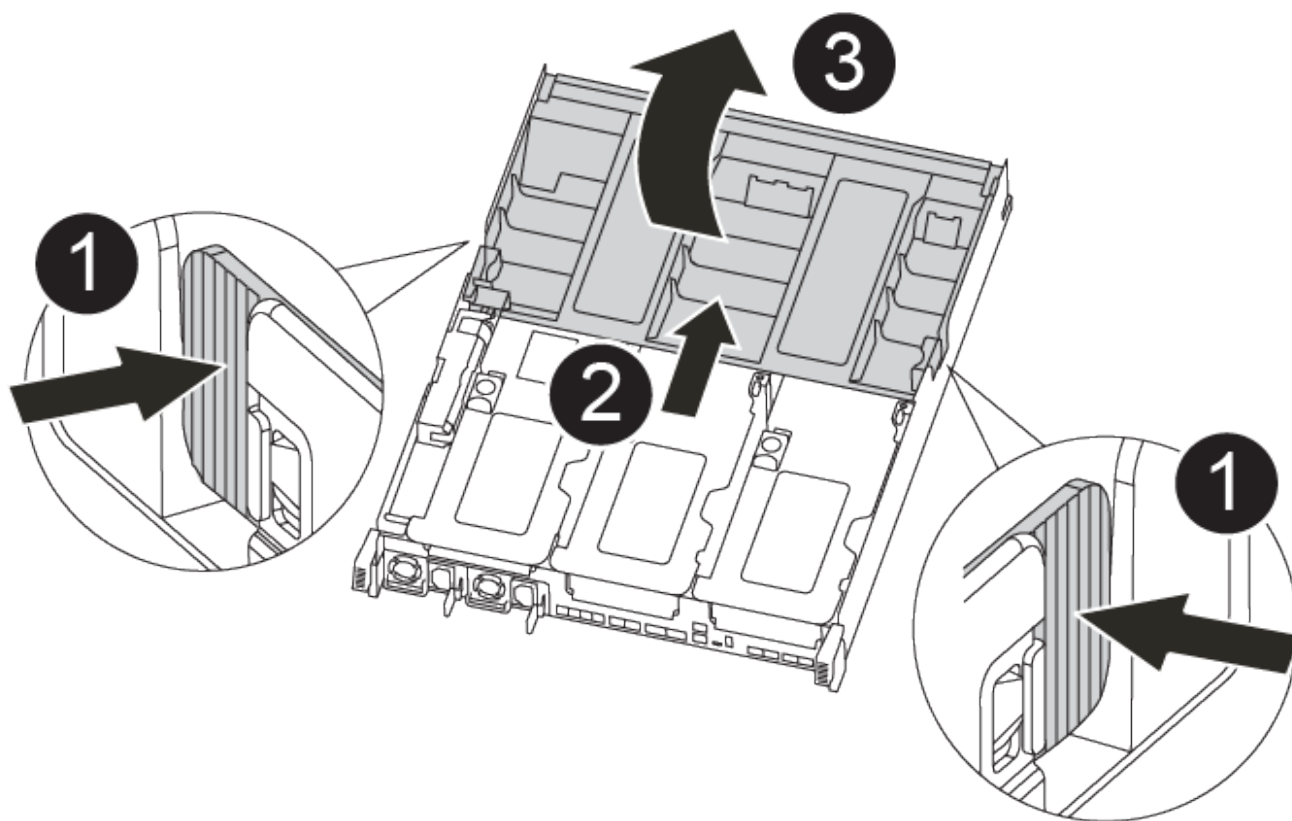
You must log into the NetApp Support Site to display the *Statement of Volatility* for your system.

You can use the following animation, illustration, or the written steps to replace the boot media.

[Animation - Replace the boot media](#)

Steps

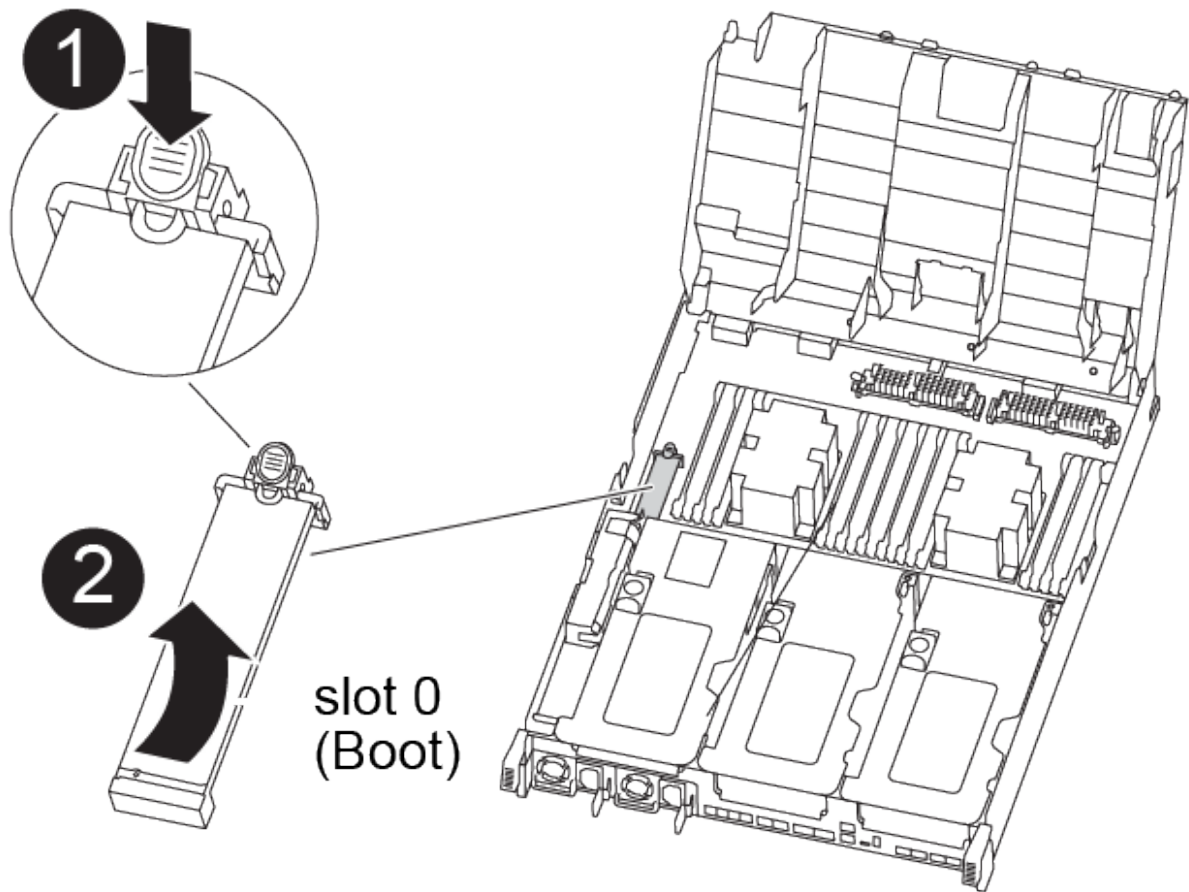
1. Open the air duct:



1	Locking tabs
2	Slide air duct toward back of controller
3	Rotate air duct up

- a. Press the locking tabs on the sides of the air duct in toward the middle of the controller module.
- b. Slide the air duct toward the back of the controller module, and then rotate it upward to its completely open position.

2. Locate and remove the boot media from the controller module:



1	Press blue button
2	Rotate boot media up and remove from socket

- a. Press the blue button at the end of the boot media until the lip on the boot media clears the blue button.
 - b. Rotate the boot media up and gently pull the boot media out of the socket.
 3. Align the edges of the replacement boot media with the boot media socket, and then gently push it into the socket.
 4. Check the boot media to make sure that it is seated squarely and completely in the socket.
- If necessary, remove the boot media and reseal it into the socket.
5. Lock the boot media in place:
 - a. Rotate the boot media down toward the motherboard.
 - b. Placing a finger at the end of the boot media by the blue button, push down on the boot media end to engage the blue locking button.
 - c. While pushing down on the boot media, lift the blue locking button to lock the boot media in place.
 6. Close the air duct.

Step 3: Transfer the boot image to the boot media

The replacement boot media that you installed does not have a boot image, so you need to transfer a boot image using a USB flash drive.

Before you begin

- You must have a USB flash drive, formatted to MBR/FAT32, with at least 4GB capacity
- A copy of the same image version of ONTAP as what the impaired controller was running. You can download the appropriate image from the Downloads section on the NetApp Support Site
 - If NVE is enabled, download the image with NetApp Volume Encryption, as indicated in the download button.
 - If NVE is not enabled, download the image without NetApp Volume Encryption, as indicated in the download button.
- If your system is an HA pair, you must have a network connection.
- If your system is a stand-alone system you do not need a network connection, but you must perform an additional reboot when restoring the `var` file system.

Steps

1. Download and copy the appropriate service image from the NetApp Support Site to the USB flash drive.
 - a. Download the service image to your work space on your laptop.
 - b. Unzip the service image.



If you are extracting the contents using Windows, do not use WinZip to extract the netboot image. Use another extraction tool, such as 7-Zip or WinRAR.

There are two folders in the unzipped service image file:

- `boot`
- `efi`

- c. Copy the `efi` folder to the top directory on the USB flash drive.



If the service image has no `efi` folder, see [EFI folder missing from Service Image download file used for boot device recovery for FAS and AFF models^](#) .

The USB flash drive should have the `efi` folder and the same Service Image (BIOS) version of what the impaired controller is running.

- d. Remove the USB flash drive from your laptop.
2. If you have not already done so, close the air duct.
 3. Align the end of the controller module with the opening in the chassis, and then gently push the controller module halfway into the system.
 4. Reinstall the cable management device and recable the system, as needed.

When recabling, remember to reinstall the media converters (SFPs or QSFPs) if they were removed.

5. Insert the USB flash drive into the USB slot on the controller module.

Make sure that you install the USB flash drive in the slot labeled for USB devices, and not in the USB console port.

6. Complete the installation of the controller module:

- a. Plug the power cord into the power supply, reinstall the power cable locking collar, and then connect the power supply to the power source.
- b. Firmly push the controller module into the chassis until it meets the midplane and is fully seated.

The locking latches rise when the controller module is fully seated.



Do not use excessive force when sliding the controller module into the chassis to avoid damaging the connectors.

- c. Rotate the locking latches upward, tilting them so that they clear the locking pins, and then lower them into the locked position.
- d. Plug the power cords into the power supplies, reinstall the power cable locking collar, and then connect the power supplies to the power source.

The controller module begins to boot as soon as power is restored. Be prepared to interrupt the boot process.

- e. If you have not already done so, reinstall the cable management device.

7. Interrupt the boot process by pressing Ctrl-C to stop at the LOADER prompt.

If you miss this message, press Ctrl-C, select the option to boot to Maintenance mode, and then `halt` the controller to boot to LOADER.

8. If the controller is in a stretch or fabric-attached MetroCluster, you must restore the FC adapter configuration:

- a. Boot to Maintenance mode: `boot_ontap maint`
- b. Set the MetroCluster ports as initiators: `ucadmin modify -m fc -t initiator adapter_name`
- c. Halt to return to Maintenance mode: `halt`

The changes will be implemented when the system is booted.

Manual boot media recovery from a USB drive - AFF A400

After installing the new boot media device in your system, you can boot the recovery image from a USB drive and restore the configuration from the partner node.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

Before you begin

- Ensure your console is connected to the impaired controller.
- Verify the e0S interface (e0M wrench port) on the impaired controller is not faulty.
- Verify you have a USB flash drive with the recovery image.

- Determine if your system uses encryption. You will need to select the appropriate option in step 3 based on whether encryption is enabled.

Steps

1. From the LOADER prompt on the impaired controller, boot the recovery image from the USB flash drive:

```
boot_recovery
```

The recovery image is downloaded from the USB flash drive.

2. When prompted, enter the name of the image or press **Enter** to accept the default image displayed in brackets.
3. Restore the var file system using the procedure for your ONTAP version:

ONTAP 9.16.0 or earlier

Complete the following steps on the impaired controller and partner controller:

- a. **On the impaired controller:** Press `Y` when you see `Do you want to restore the backup configuration now?`
- b. **On the impaired controller:** If prompted, press `Y` to overwrite `/etc/ssh/ssh_host_ecdsa_key`.
- c. **On the partner controller:** Set the impaired controller to advanced privilege level:

```
set -privilege advanced
```

- d. **On the partner controller:** Run the restore backup command:

```
system node restore-backup -node local -target-address  
impaired_node_IP_address
```



If you see any message other than a successful restore, contact NetApp Support.

- e. **On the partner controller:** Return to admin level:

```
set -privilege admin
```

- f. **On the impaired controller:** Press `Y` when you see `Was the restore backup procedure successful?`
- g. **On the impaired controller:** Press `Y` when you see `...would you like to use this restored copy now?`
- h. **On the impaired controller:** Press `Y` when prompted to reboot, then press `Ctrl-C` when you see the Boot Menu.
- i. **On the impaired controller:** Do one of the following:
 - If the system does not use encryption, select *Option 1 Normal Boot* from the Boot Menu.
 - If the system uses encryption, go to [Restore encryption](#).

ONTAP 9.16.1 or later

Complete the following steps on the impaired controller:

- a. Press `Y` when prompted to restore the backup configuration.

After the restore procedure is successful, this message displays: `syncflash_partner:`
`Restore from partner complete`

- b. Press `Y` when prompted to confirm that the restore backup was successful.
- c. Press `Y` when prompted to use the restored configuration.
- d. Press `Y` when prompted to reboot the node.
- e. Press `Y` when prompted to reboot again, then press `Ctrl-C` when you see the Boot Menu.
- f. Do one of the following:
 - If the system does not use encryption, select *Option 1 Normal Boot* from the Boot Menu.

- If the system uses encryption, go to [Restore encryption](#).

4. Connect the console cable to the partner controller.
5. Return the controller to normal operation by giving back its storage:

```
storage failover giveback -fromnode local
```

6. If you disabled automatic giveback, reenable it:

```
storage failover modify -node local -auto-giveback true
```

7. If AutoSupport is enabled, restore automatic case creation:

```
system node autosupport invoke -node * -type all -message MAINT=END
```

Restore encryption - AFF A400

Restore encryption on the replacement boot media.

If your storage system is running ONTAP 9.17.1 or later, use the [automated boot recovery procedure](#). If your system is running an earlier version of ONTAP, you must use the manual boot recovery process.

Complete the appropriate steps to restore encryption on your system based on your key manager type. If you are unsure which key manager your system uses, check the settings you captured at the beginning of the boot media replacement procedure.

Onboard Key Manager (OKM)

Restore the Onboard Key Manager (OKM) configuration from the ONTAP boot menu.

Before you begin

Ensure you have the following information available:

- Cluster-wide passphrase entered while [enabling onboard key management](#)
- [Backup information for the Onboard Key Manager](#)
- Verification that you have the correct passphrase and backup data using the [How to verify onboard key management backup and cluster-wide passphrase](#) procedure

Steps

On the impaired controller:

1. Connect the console cable to the impaired controller.
2. From the ONTAP boot menu, select the appropriate option:

ONTAP version	Select this option
ONTAP 9.8 or later	Select option 10. Show example boot menu Please choose one of the following: (1) Normal Boot. (2) Boot without /etc/rc. (3) Change password. (4) Clean configuration and initialize all disks. (5) Maintenance mode boot. (6) Update flash from backup config. (7) Install new software first. (8) Reboot node. (9) Configure Advanced Drive Partitioning. (10) Set Onboard Key Manager recovery secrets. (11) Configure node for external key management. Selection (1-11)? 10

ONTAP version	Select this option
ONTAP 9.7 and earlier	Select the hidden option <code>recover_onboard_keymanager</code> Show example boot menu Please choose one of the following: (1) Normal Boot. (2) Boot without <code>/etc/rc</code>. (3) Change password. (4) Clean configuration and initialize all disks. (5) Maintenance mode boot. (6) Update flash from backup config. (7) Install new software first. (8) Reboot node. (9) Configure Advanced Drive Partitioning. Selection (1-19)? <code>recover_onboard_keymanager</code>

3. Confirm that you want to continue the recovery process when prompted:

Show example prompt

```
This option must be used only in disaster recovery procedures. Are you
sure? (y or n):
```

4. Enter the cluster-wide passphrase twice.

While entering the passphrase, the console does not show any input.

Show example prompt

```
Enter the passphrase for onboard key management:

Enter the passphrase again to confirm:
```

5. Enter the backup information:

- a. Paste the entire content from the BEGIN BACKUP line through the END BACKUP line, including the dashes.

Show example prompt

Enter the backup data:

[illegible]

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA
01234567890123456789012345678901234567890123456789012345678901
23
12345678901234567890123456789012345678901234567890123456789012
34
23456789012345678901234567890123456789012345678901234567890123
45
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AA

-----END
BACKUP-----
```

b. Press Enter twice at the end of the input.

The recovery process completes and displays the following message:

Successfully recovered keymanager secrets.

Show example prompt

```
Trying to recover keymanager secrets....
Setting recovery material for the onboard key manager
Recovery secrets set successfully
Trying to delete any existing km_onboard.wkeydb file.

Successfully recovered keymanager secrets.

*****
*****
* Select option "(1) Normal Boot." to complete recovery
process.
*
* Run the "security key-manager onboard sync" command to
synchronize the key database after the node reboots.
*****
*****
```



Do not proceed if the displayed output is anything other than Successfully recovered keymanager secrets. Perform troubleshooting to correct the error.

6. Select option 1 from the boot menu to continue booting into ONTAP.

Show example prompt

```
*****
*****
* Select option "(1) Normal Boot." to complete the recovery
process.
*
*****
*****

(1) Normal Boot.
(2) Boot without /etc/rc.
(3) Change password.
(4) Clean configuration and initialize all disks.
(5) Maintenance mode boot.
(6) Update flash from backup config.
(7) Install new software first.
(8) Reboot node.
(9) Configure Advanced Drive Partitioning.
(10) Set Onboard Key Manager recovery secrets.
(11) Configure node for external key management.
Selection (1-11)? 1
```

7. Confirm that the controller's console displays the following message:

```
Waiting for giveback...(Press Ctrl-C to abort wait)
```

On the partner controller:

8. Giveback the impaired controller:

```
storage failover giveback -fromnode local -only-cfo-aggregates true
```

On the impaired controller:

9. After booting with only the CFO aggregate, synchronize the key manager:

```
security key-manager onboard sync
```

10. Enter the cluster-wide passphrase for the Onboard Key Manager when prompted.

Show example prompt

Enter the cluster-wide passphrase for the Onboard Key Manager:

All offline encrypted volumes will be brought online and the corresponding volume encryption keys (VEKs) will be restored automatically within 10 minutes. If any offline encrypted volumes are not brought online automatically, they can be brought online manually using the "volume online -vserver <vserver> -volume <volume_name>" command.



If the sync is successful, the cluster prompt is returned with no additional messages. If the sync fails, an error message appears before returning to the cluster prompt. Do not continue until the error is corrected and the sync runs successfully.

11. Verify that all keys are synced:

```
security key-manager key query -restored false
```

The command should return no results. If any results appear, repeat the sync command until no results are returned.

On the partner controller:

12. Giveback the impaired controller:

```
storage failover giveback -fromnode local
```

13. Restore automatic giveback if you disabled it:

```
storage failover modify -node local -auto-giveback true
```

14. If AutoSupport is enabled, restore automatic case creation:

```
system node autosupport invoke -node * -type all -message MAINT=END
```

External Key Manager (EKM)

Restore the External Key Manager configuration from the ONTAP boot menu.

Before you begin

Gather the following files from another cluster node or from your backup:

- /cfcard/knip/servers.cfg file or the KMIP server address and port
- /cfcard/knip/certs/client.crt file (client certificate)
- /cfcard/knip/certs/client.key file (client key)
- /cfcard/knip/certs/CA.pem file (KMIP server CA certificates)

Steps

On the impaired controller:

1. Connect the console cable to the impaired controller.
2. Select option 11 from the ONTAP boot menu.

Show example boot menu

```
(1) Normal Boot.
(2) Boot without /etc/rc.
(3) Change password.
(4) Clean configuration and initialize all disks.
(5) Maintenance mode boot.
(6) Update flash from backup config.
(7) Install new software first.
(8) Reboot node.
(9) Configure Advanced Drive Partitioning.
(10) Set Onboard Key Manager recovery secrets.
(11) Configure node for external key management.
Selection (1-11)? 11
```

3. Confirm you have gathered the required information when prompted:

Show example prompt

```
Do you have a copy of the /cfcard/kmip/certs/client.crt file?
{y/n}
Do you have a copy of the /cfcard/kmip/certs/client.key file?
{y/n}
Do you have a copy of the /cfcard/kmip/certs/CA.pem file? {y/n}
Do you have a copy of the /cfcard/kmip/servers.cfg file? {y/n}
```

4. Enter the client and server information when prompted:
 - a. Enter the client certificate (client.crt) file contents, including the BEGIN and END lines.
 - b. Enter the client key (client.key) file contents, including the BEGIN and END lines.
 - c. Enter the KMIP server CA(s) (CA.pem) file contents, including the BEGIN and END lines.
 - d. Enter the KMIP server IP address.
 - e. Enter the KMIP server port (press Enter to use the default port 5696).

Show example

```
Enter the client certificate (client.crt) file contents:
-----BEGIN CERTIFICATE-----
<certificate_value>
-----END CERTIFICATE-----

Enter the client key (client.key) file contents:
-----BEGIN RSA PRIVATE KEY-----
<key_value>
-----END RSA PRIVATE KEY-----

Enter the KMIP server CA(s) (CA.pem) file contents:
-----BEGIN CERTIFICATE-----
<certificate_value>
-----END CERTIFICATE-----

Enter the IP address for the KMIP server: 10.10.10.10
Enter the port for the KMIP server [5696]:

System is ready to utilize external key manager(s).
Trying to recover keys from key servers....
kmip_init: configuring ports
Running command '/sbin/ifconfig e0M'
..
..
kmip_init: cmd: ReleaseExtraBSDPort e0M
```

The recovery process completes and displays the following message:

Successfully recovered keymanager secrets.

Show example

```
System is ready to utilize external key manager(s).
Trying to recover keys from key servers....
Performing initialization of OpenSSL
Successfully recovered keymanager secrets.
```

5. Select option 1 from the boot menu to continue booting into ONTAP.

Show example prompt

```
*****
*****
* Select option "(1) Normal Boot." to complete the recovery
process.
*
*****
*****

(1) Normal Boot.
(2) Boot without /etc/rc.
(3) Change password.
(4) Clean configuration and initialize all disks.
(5) Maintenance mode boot.
(6) Update flash from backup config.
(7) Install new software first.
(8) Reboot node.
(9) Configure Advanced Drive Partitioning.
(10) Set Onboard Key Manager recovery secrets.
(11) Configure node for external key management.
Selection (1-11)? 1
```

6. Restore automatic giveback if you disabled it:

```
storage failover modify -node local -auto-giveback true
```

7. If AutoSupport is enabled, restore automatic case creation:

```
system node autosupport invoke -node * -type all -message MAINT=END
```

Return the failed boot media to NetApp - AFF A400

Return the failed part to NetApp, as described in the RMA instructions shipped with the kit. See the [Part Return and Replacements](#) page for further information.

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